RASHMI DANGOL

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CAREER OBJECTIVE

Innovative Digital Agronomist and FAA-certified Remote Pilot with experience in environmental data collection, geospatial technologies, and remote sensing applications. Skilled in field-based research, GIS, spatial data analysis, and scientific reporting. Proven track record at **Syngenta** and **Bayer Crop Science** in optimizing product placement, high-throughput phenotyping, and automated field mapping to enhance crop/pasture performance.

TECHNICAL SKILLS

- **Programming & Data Science:** Python (ArcPy, GeoPandas), Computer Vision (YOLO), Deep Learning, SAS, SPSS, Microsoft Power BI.
- **Field Operations:** DJI Matrice 300 RTK, GPS Rover, High-Throughput Phenotyping, Soil Sampling, Herbicide Application
- **Industry Tools:** Emlid Flow**,** Smart Farm Plus, ArcGIS Survey123, Self-Service Shape, KoboToolbox, Field Mapping.
- **GIS & Remote Sensing:** ArcGIS Pro, ArcGIS Online, ENVI, Pix4D Mapper, ArcGIS Experience Builder, ArcGIS StoryMaps, ArcGIS Dashboard, Satellite Imagery Analysis.

PROFESSIONAL EXPERIENCE

Product Placement Phenotyping Intern | **Syngenta,** Malta, IL | **May 2025 – Aug 2025**

- **Geospatial Field Mapping:** Engineered digital field maps using Google Earth Pro and Self-Service Shape for precise project planning and spatial analysis.
- **UAV Operations:** Led flight operations using DJI Matrice 300 RTK drone to capture high resolution imagery for early-season vigor and corn emergence quality.
- **Data Pipeline Development:** Collaborated on a Python-based machine-learning image processing pipeline to generate first-model estimators for crop health indices.
- **Insight Generation:** Delivered actionable geospatial insights to the product placement team to optimize variety selection based on spatial variability.

Research Assistant (Geospatial Focused) | **Texas A&M University** | **Illinois State University | May 2022 – Present**

- **Environmental and Field Research:** Conducted field-based soil investigations involving soil sampling, and data collection to evaluate soil organic matter variability across different Illinois cropping systems.
- **Data Management and Analysis**: Maintained accurate field records and supported environmental assessment through data analysis using Python.
- **Remote Sensing Research and AI Integration:** Conducted large-scale experiments utilizing UAV imagery to identify invasive weed species to evaluate pastureland management.
- **Workflow Automation**: Created custom Python functions using ArcPy to automate geospatial data processing, increasing efficiency in time-series analysis.
- **Flood Analysis & Geospatial Analysis:** Conducted flood analysis for Kerr County using soil volumetric content and land cover data in **ArcGIS Pro** and **ArcGIS Notebooks**.
- **Technical Reports and Communication**: Prepared project presentations, research reports, maps, and data summaries communicating scientific findings to technical and non-technical audiences.

Seed Technology Intern | **Bayer Crop Science**, Oxnard, CA | **May 2023 – Aug 2023**

- **Yield Analysis:** Executed field data collection for yield performance using GPS rovers, ensuring high data integrity for the LATAM seeds technical sheets.
- **Data Coordination:** Managed soil analysis and tissue culture data via the Smart Farm Plus application to support seed quality protocols.

Instructor | **Shree Janata Secondary School,** Nepal | **Dec 2019 – Apr 2022**

- **Education and Curriculum Development:** Taught soil science and organic farming courses to higher secondary students. Collaborated with a multidisciplinary team to develop individualized education plans for higher secondary level students with diverse learning needs.
- **Student Assessment & Problem Solving:** Developed strong organizational and problem-solving skills by identifying and resolving student learning gaps.

Digital Enumerator/Surveyor | **CIMMYT**, Nepal | **Jun 2018 – Aug 2018**

- **Survey Leadership and Data Collection:** Led a 6-member team for the National Seed and Fertilizer Project (NSAF), leveraging KoboToolbox to digitize data collection and evaluate the effectiveness of ICT tools among rural smallholder farmers.

EDUCATION

- **Ph.D. in Precision Agronomy** | Texas A&M University, College Station, TX | *May* 2027
- **Graduate Certificate in GIS** | Texas A&M University, College Station, TX | *Dec* 2025
- **M.S. in Applied Science** | Illinois State University, Normal, IL | *May 2024*
- **Bachelor of Science in Agriculture** | Agriculture and Forestry University, Nepal | *Sep 2019*

CERTIFICATIONS & AWARDS

- **FAA Part 107 Certified Remote Pilot (**Commercial Drone License) | *Sep 2024*
- **Visualizing Data with Python** | IBM | *Jun 2025*
- **James L. Fisher Outstanding Thesis Award** | Illinois State University | *Nov 2024*
- **Outstanding Graduate Student Recognition** | Illinois State University | *Apr 2024*